

Chi-Square Test & Multinomial Distribution

L-10

Page 1: Multinomial Setup and MLE

Vector Notation:

$$\underline{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}, \quad \sum_{i=1}^k x_i = n$$

Multinomial Distribution

Let $\mathbf{X} \sim \text{Multinomial}(n, \mathbf{p})$, where $p_i \geq 0$ and $\sum_{i=1}^k p_i = 1$. Here, X_i represents the number of times the i -th outcome happens in n trials.

The Probability Mass Function (PMF) is given by:

$$P(X_1 = x_1, \dots, X_k = x_k) = \begin{cases} \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k} & \text{if } \sum_{i=1}^k x_i = n \\ 0 & \text{otherwise} \end{cases}$$

Estimation (MLE)

- Maximum Likelihood Estimator: $\hat{p}_{\text{MLE}} = \frac{1}{n} \mathbf{X}$
- Expectation: $E[\hat{p}_{\text{MLE}}] = \mathbf{p}$ (Unbiased)
- Variance/Covariance: $\text{Var}(\hat{p}) = \frac{1}{n} \Sigma$

The elements of the covariance matrix Σ are:

$$\sigma_{ii} = p_i(1 - p_i) \quad \text{and} \quad \sigma_{ij} = -p_i p_j \quad (\text{for } i \neq j)$$

Asymptotic Properties

By the Strong Law of Large Numbers (SLLN):

$$\hat{p} \xrightarrow{a.s.} p$$

By the Central Limit Theorem (CLT):

$$\frac{\sqrt{n}(\hat{p}_i - p_i)}{\sqrt{p_i(1 - p_i)}} \Rightarrow N(0, 1)$$

$$\forall i = 1, 2, \dots, k$$

Hypothesis Test:

$$H_0 : \mathbf{p} = \mathbf{p}_0 \quad \text{vs} \quad H_a : \mathbf{p} \neq \mathbf{p}_0$$

We can represent the count vector as a sum of indicator vectors: $\mathbf{X} = \sum_{j=1}^n \mathbf{Y}_j$, where $\mathbf{Y}_j$ is a standard basis vector (e.g., $[0, 1, 0 \dots 0]^T$) indicating the outcome of the j -th trial.

Multivariate CLT and Singularity

Moments of Indicator Variables

Since $\mathbf{Y}_j$ are independent and identically distributed (i.i.d):

$$E[\mathbf{Y}_j] = \mathbf{p}, \quad \text{Var}(\mathbf{Y}_j) = \Sigma$$

Multivariate CLT

$$\sqrt{n}\Sigma^{-1/2} \left(\frac{\mathbf{X}}{n} - \mathbf{p} \right) \Rightarrow N(\mathbf{0}, I)$$

Issue: Singularity

The covariance matrix Σ is singular.

$$\Sigma = \begin{pmatrix} p_1(1-p_1) & -p_1p_2 \\ -p_1p_2 & p_2(1-p_2) \end{pmatrix} \quad (\text{Example for } k=2?)$$

Property: $\mathbf{1}^T \Sigma = \mathbf{0}^T$ (Rows/Columns sum to 0).

Reduction (Handling Singularity)

To resolve singularity, we drop the last component k . Define reduced vectors:

$$\mathbf{p}^* = (p_1, \dots, p_{k-1})^T, \quad \mathbf{X}^* = (X_1, \dots, X_{k-1})^T$$

Check linear dependence:

$$\sum_{j=1}^k p_j = 1 \implies p_k = 1 - \sum_{i=1}^{k-1} p_i$$

Checking covariance row sum (for index 2 as an example):

$$-p_1p_2 - p_2^2 - \dots + p_2(1-p_2) = -p_2\left(\sum_{all} p_i\right) + p_2 = -p_2(1) + p_2 = 0$$

Reduced Hypothesis:

$$H_0 : \mathbf{p}^* = \mathbf{p}_0^*$$

Test Statistic components:

$$X_k = n - \sum_{i=1}^{k-1} X_i$$

Properties of reduced vector:

$$E[\mathbf{X}^*] = \mathbf{p}^*, \quad \text{Var}(\mathbf{X}^*) = \Sigma^*$$

The reduced vector follows a non-singular Multivariate Normal asymptotically:

$$\sqrt{n}\left(\frac{\mathbf{X}^*}{n} - \mathbf{p}^*\right) \Rightarrow N_{k-1}(\mathbf{0}, \Sigma)$$

Let $\mathbf{Z}_n^*$ be the standardized version. Then the quadratic form:

$$\mathbf{Z}^{*T} \mathbf{Z}^* \Rightarrow \chi_{k-1}^2$$

Page 3: Inverting the Covariance Matrix

We examine the quadratic form:

$$n \left(\frac{\mathbf{X}^*}{n} - \mathbf{p}^* \right)^T (\Sigma^*)^{-1} \left(\frac{\mathbf{X}^*}{n} - \mathbf{p}^* \right) \Rightarrow \chi_{k-1}^2$$

Claim: The inverse of the reduced covariance matrix is:

$$(\Sigma^*)^{-1} = \text{diag} \left(\frac{1}{\mathbf{p}^*} \right) + \frac{1}{p_k} \mathbf{1} \mathbf{1}^T$$

Let $A = (\Sigma^*)^{-1}$. We must show $A\Sigma^* = \Sigma^*A = I$.

Recall:

$$\Sigma^* = \text{diag}(\mathbf{p}^*) - \mathbf{p}^* \mathbf{p}^{*T}$$

Let's compute the product $A\Sigma^*$:

$$A\Sigma^* = \left[\text{diag} \left(\frac{1}{\mathbf{p}^*} \right) + \frac{1}{p_k} \mathbf{1} \mathbf{1}^T \right] [\text{diag}(\mathbf{p}^*) - \mathbf{p}^* \mathbf{p}^{*T}]$$

Expanding the terms:

$$\begin{aligned} A\Sigma^* &= \underbrace{\text{diag} \left(\frac{1}{\mathbf{p}^*} \right) \text{diag}(\mathbf{p}^*)}_I - \underbrace{\text{diag} \left(\frac{1}{\mathbf{p}^*} \right) \mathbf{p}^* \mathbf{p}^{*T}}_{\mathbf{1} \mathbf{p}^{*T}} \\ &\quad + \underbrace{\frac{1}{p_k} \mathbf{1} \mathbf{1}^T \text{diag}(\mathbf{p}^*)}_{\frac{1}{p_k} \mathbf{1} \mathbf{p}^{*T}} - \underbrace{\frac{1}{p_k} \mathbf{1} \mathbf{1}^T \mathbf{p}^* \mathbf{p}^{*T}}_{\frac{1}{p_k} \mathbf{1} (\mathbf{1}^T \mathbf{p}^*) \mathbf{p}^{*T}} \end{aligned}$$

Note that $\mathbf{1}^T \mathbf{p}^* = \sum_{i=1}^{k-1} p_i = 1 - p_k$. Substituting this back:

$$\begin{aligned} A\Sigma^* &= I - \mathbf{1} \mathbf{p}^{*T} + \frac{1}{p_k} \mathbf{1} \mathbf{p}^{*T} - \frac{1}{p_k} \mathbf{1} (1 - p_k) \mathbf{p}^{*T} \\ &= I - \mathbf{1} \mathbf{p}^{*T} + \frac{1}{p_k} \mathbf{1} \mathbf{p}^{*T} - \left(\frac{1}{p_k} - 1 \right) \mathbf{1} \mathbf{p}^{*T} \\ &= I + \mathbf{1} \mathbf{p}^{*T} \left(-1 + \frac{1}{p_k} - \frac{1}{p_k} + 1 \right) \\ &= I + \mathbf{1} \mathbf{p}^{*T} (0) \\ &= I \end{aligned}$$

Thus, the inverse is verified.

Test of Hypothesis $H_0 : \mathbf{p} = \mathbf{p}_0$ Under H_0 , the statistic is:

$$B_n = n \left(\frac{\mathbf{X}^*}{n} - \mathbf{p}_0^* \right)^T A \left(\frac{\mathbf{X}^*}{n} - \mathbf{p}_0^* \right) \Rightarrow \chi_{k-1}^2$$

where $A = \text{diag}\left(\frac{1}{\mathbf{p}_0^*}\right) + \frac{1}{p_{0k}} \mathbf{1}\mathbf{1}^T$.

Pearson's Chi-Square Statistic

The Statistic B_n

We substitute the matrix A back into the quadratic form.

$$A = \text{diag}\left(\frac{1}{\mathbf{p}_0^*}\right) + \frac{1}{p_{0k}} \mathbf{1}\mathbf{1}^T$$

Decision Rule:

- B_n is the test statistic.
- Reject H_0 according to quantiles of χ_{k-1}^2 .
- Rejection region is right-tailed (Reject for large values of B_n).

Claim 2: Equivalence to Pearson's Formula

We can show that the matrix form simplifies to the summation form:

$$B_n = \sum_{i=1}^k \frac{(x_i - np_i)^2}{np_i} = \sum_{i=1}^k \frac{(\text{Obs}_i - \text{Exp}_i)^2}{\text{Exp}_i}$$

where:

- k : number of cells.
- x_i : Observed count (Obs_i).
- np_i : Expected count (Exp_i).

This result relies on the Multivariate Central Limit Theorem (MCLT).